

Multiple Choice (10 marks)

1. What are the results of fertilization?
 - (a) DNA replication, cell division, energy production, and protein synthesis
 - (b) Photosynthesis, respiration, transpiration, and nutrient absorption
 - (c) Synthesis of hormones, enzymatic reactions, metabolic activity, and homeostasis
 - (d) Evolution, mutation, adaptation, and natural selection
 - (e) Formation of a diploid cell, development of multicellular organism, growth, and aging
2. Our eyes are the principal organs involved in seeing. Yet we are able to perform manual acts such as dressing or tying knots when our eyes are closed. Why?
 - (a) Gustatory receptors
 - (b) Photoreceptors
 - (c) Auditory receptors
 - (d) Proprioceptors
 - (e) Thermoreceptors
3. A change in a neuron membrane potential from +50 millivolts to -70 millivolts is considered
 - (a) a threshold potential
 - (b) hyperpolarization
 - (c) an action potential
 - (d) an equilibrium potential
 - (e) a resting potential
4. Discuss some of the elements found in advanced animal societies.
 - (a) Language development, agricultural practices, and written communication are elements found in advanced animal societies.
 - (b) Economic systems, legal structures, and political elections are elements found in advanced animal societies.
 - (c) Adaptation, evolution and species diversity are elements found in advanced animal societies.
 - (d) Cultural traditions, artistic expression, and philosophy are elements found in advanced animal societies.

- (e) Leadership, territory and dominance hierarchy are elements found in advanced animal societies.
5. Which of the following is not known to be involved in the control of cell division?
- (a) Microtubules
 - (b) Checkpoints
 - (c) DNA polymerase
 - (d) Centrosomes
 - (e) Cyclins

True / False (10 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'Why does milk \{\}\textquotedblleftspoil\{\}\textquotedblright when kept in a refrigerator?' is 'Thermophiles cause milk to spoil in the refrigerator'.
2. True or False: The correct answer to 'All of the following are examples of events that can prevent interspecific breeding EXCEPT' is 'the potential mates have similar breeding seasons'.
3. True or False: The correct answer to 'We know that radiation and chemicals can cause cancer. Can viruses be a cause of cancer?' is 'Only inherited genetics can cause cancer'.
4. True or False: The correct answer to 'What are the different types of point mutations?' is 'Base substitutions and small insertions and deletions'.
5. True or False: The correct answer to 'A woman opens a can of ham and beans and suspects food poisoning . She boils the contents for half an hour. A few hours after eating the...' is 'The woman was poisoned by Clostridium botulinum toxin.'.

Long Form Response (20 marks)

Provide thorough reasoning and reference key steps.

1. A newborn infant is examined by a doctor, who determines that the child has syphilis. Explain how the child contracted syphilis .
2. The polytene chromosomes seen in *Drosophila silvarentis* with the aid of a light microscope are

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